

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-II]

TEST # 01

DATE : 16-12-2019

REVISION TEST

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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SPACE FOR ROUGH WORK

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BEWARE OF NEGATIVE MARKING

1. $\cos(120^\circ)$ equals to:

(1) $\frac{1}{2}$ (2) $-\frac{1}{2}$

(3) $\frac{\sqrt{3}}{2}$ (4) $-\frac{\sqrt{3}}{2}$

2. Value of $(\sin^2 37^\circ - 1) + (\cos^2 37^\circ - 1) = \dots$

(1) 1 (2) -1

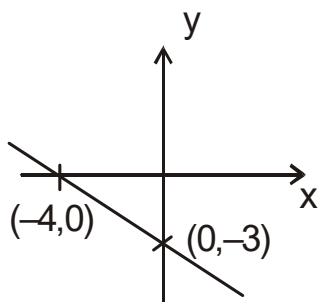
(3) 2 (4) -2

3. Of $y = x + \frac{1}{x}$, then $\frac{dy}{dx}$ is

(1) $1 + \frac{1}{x^2}$ (2) $x - \frac{1}{x^2}$

(3) $1 - \frac{1}{x^2}$ (4) $x^2 - \frac{1}{x^2}$

4. Find slope of straight line:



(1) $\frac{4}{3}$ (2) $\frac{3}{4}$

(3) $-\frac{4}{3}$ (4) $-\frac{3}{4}$

5. If $f = x^2yz + xz$. Find $\frac{\partial f}{\partial x}$ at (1, 2, 1) is

(1) 5 (2) 4
(3) 3 (4) Zero

6. $1 + \tan^2 4\theta$ is equal to

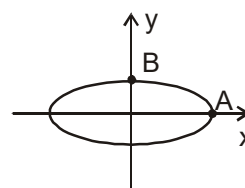
(1) $\sec^2 \theta$ (2) $4\sec^2 \theta$
(3) $\sec^2 4\theta$ (4) $4 \sec^2 4\theta$

7. If $\sin \theta = -\frac{1}{3}$, then $\cos \theta$ will be -

(1) $\pm \frac{8}{9}$ (2) $\pm \frac{4}{3}$

(3) $\pm \frac{2\sqrt{2}}{3}$ (4) $\pm \frac{3}{4}$

8. For ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$, the co-ordinates of points A and B are respectively:



(1) (0, 4) and (5, 0)
(2) (4, 0) and (0, 5)
(3) (5, 0) and (0, 4)
(4) (25, 0) and (0, 16)

9. Find sum of given arithmetic progression:
 $1+4+7+\dots+94$

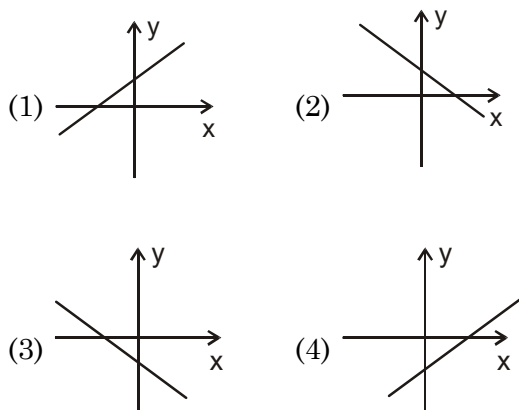
(1) 31 (2) 106
(3) 1519 (4) 1520

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10. Sum of $1 + \frac{1}{9} + \frac{1}{81} + \dots \infty$ is:

- (1) $\frac{8}{9}$ (2) $\frac{2}{3}$
(3) $\frac{9}{8}$ (4) $\frac{3}{2}$

11. Correct graph for $y = 5x - 2$ is



12. Determine the average value of $y = 4x + 5$ in the interval $0 \leq x \leq 2$

- (1) 13 (2) 5
(3) 9 (4) 6

13. $\int_3^2 \frac{(x^2 - 1)}{x + 1} dx$

- (1) $\frac{1}{2}$ (2) $-\frac{1}{2}$
(3) $\frac{3}{2}$ (4) $-\frac{3}{2}$

14. If $y = e^x \cdot \cot x$ then $\frac{dy}{dx}$ will be :

- (1) $e^x \cot x - \operatorname{cosec}^2 x$
(2) $e^x \operatorname{cosec}^2 x$
(3) $e^x [\cot x - \operatorname{cosec}^2 x]$
(4) $e^x \cot x$

15. If $y = e^{(x^2+x)}$ then $\frac{dy}{dx} = ?$

- (1) e^{x^2+x} (2) $e^{(2x+1)}$
(3) $(2x+1)e^{(x^2+x)}$ (4) 0

16. What is the maximum number of rectangular components into which a vector can be split in space ?

- (1) 2 (2) 3
(3) 4 (4) Infinite

17. The vectors $\vec{A}$ and $\vec{B}$ are such that $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$ then the angle between the

two vectors $\vec{A}$ and $\vec{B}$ will be :

- (1) 0° (2) 60°
(3) 90° (4) 180°

18. If $\vec{A} \cdot \vec{B} = |\vec{A} \times \vec{B}|$, then angle θ between vectors $\vec{A}$ and $\vec{B}$ is :

- (1) 0 (2) $\frac{\pi}{4}$
(3) $\frac{\pi}{2}$ (4) π

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19. If $\vec{A} = \vec{B} + \vec{C}$ and the magnitudes of $\vec{A}$, $\vec{B}$ and $\vec{C}$ are 5, 4 and 3 units respectively, the angle between $\vec{A}$ and $\vec{C}$ is :

- (1) $\cos^{-1}\left(\frac{3}{5}\right)$ (2) $\cos^{-1}\left(\frac{4}{5}\right)$
(3) $\frac{\pi}{2}$ (4) $\sin^{-1}\left(\frac{3}{4}\right)$

20. The vector sum of the forces of 10 newton and 6 newton can be :

- (1) 2N (2) 8N (3) 18N (4) 20N

21. Let $\vec{A} = \hat{i}A\cos\theta + \hat{j}A\sin\theta$ be any vector.

Another vector $\vec{B}$ which is normal to $\vec{A}$ is :

- (1) $\hat{i}B\cos\theta + \hat{j}B\sin\theta$ (2) $\hat{i}B\sin\theta + \hat{j}B\cos\theta$
(3) $\hat{i}B\sin\theta - \hat{j}B\cos\theta$ (4) $\hat{i}A\cos\theta - \hat{j}A\sin\theta$

22. If $\vec{A} \times \vec{B} = 0$ and $\vec{A} \cdot \vec{B} = -AB$, then angle between $\vec{A}$ and $\vec{B}$ is ?

- (1) zero (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{2}$ (4) π

23. If $|\vec{A} \times \vec{B}| = \sqrt{3}\vec{A} \cdot \vec{B}$, then the value of $|\vec{A} + \vec{B}|$ is:

- (1) $(A^2 + B^2 + AB)^{1/2}$
(2) $\left(A^2 + B^2 + \frac{AB}{\sqrt{3}}\right)^{1/2}$
(3) $(A + B)$
(4) $(A^2 + B^2 + \sqrt{3}AB)^{1/2}$

24. The diagonals of a parallelogram are represented by vectors $\vec{P} = 5\hat{i} - 4\hat{j} + 3\hat{k}$ and $\vec{Q} = 3\hat{i} + 2\hat{j} - \hat{k}$. Then the area of the parallelogram is :

- (1) $\sqrt{171}$ unit (2) $\sqrt{72}$ unit
(3) 171 unit (4) 72 unit

25. If a vector $2\hat{i} + 3\hat{j} + 8\hat{k}$ is perpendicular to the vector $4\hat{j} - 4\hat{i} + \alpha\hat{k}$, then the value of α is :

- (1) $\frac{1}{2}$ (2) $-\frac{1}{2}$ (3) 1 (4) -1

26. Dimension of gravitational constant are :-

- (1) $[ML^2T^2]$ (2) $[M^1L^3T^{-2}]$
(3) $[M^0L^3T^2]$ (4) $[M^{-1}L^3T^{-2}]$

27. The velocity v (in cm/sec) of a particle is given in terms of time t (in sec) by the equation

$$v = at + \frac{b}{t + c}$$

The dimensions of a , b and c are :-

- | a | b | c |
|-----------------|--------|----------|
| (1) $[L^2]$ | $[T]$ | $[LT^2]$ |
| (2) $[LT^2]$ | $[LT]$ | $[L]$ |
| (3) $[LT^{-2}]$ | $[L]$ | $[T]$ |
| (4) $[L]$ | $[LT]$ | $[T^2]$ |

28. In the relation :-

$$y = a \sin(\omega t - Kx)$$

the dimensional formula for K is :

- (1) $[M^0LT]$ (2) $[M^0L^{-1}T^0]$
(3) $[M^0LT^{-1}]$ (4) $[M^0L^{-1}T^{-1}]$

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- 29.** If the speed of light (c), acceleration due to gravity (g) and pressure (p) are taken as fundamental units, the dimensions of gravitational constant (G) are :
- (1) $c^0 g p^{-3}$ (2) $c^2 g^3 p^{-2}$
(3) $c^0 g^2 p^{-1}$ (4) $c^2 g^2 p^{-2}$
- 30.** If the velocity of light (c), gravitational constant (G) and Planck's constant (h) are chosen as fundamental units, then which of the following represents the dimensions of the mass ?
- (1) $[c^{1/2} G^{1/2} h^{1/2}]$ (2) $[c^{1/2} G^{-1/2} h^{-1/2}]$
(3) $[c^{1/2} G^{-1/2} h^{1/2}]$ (4) $[c^{-1/2} G^{1/2} h^{-1/2}]$
- 31.** A student measured the diameter of a wire using a screw gauge with least count 0.001 cm and listed the measurements. The correct measurement is :
- (1) 5.320 cm (2) 5.3 cm
(3) 5.32 cm (4) 5.3200 cm
- 32.** An experiment measured quantities a , b , c and then x calculated from $x = \frac{ab^2}{c^3}$. If the percentage errors a , b , c are $\pm 1\%$, $\pm 3\%$ and $\pm 2\%$ respectively, the percentage error in x can be:
- (1) $\pm 13\%$ (2) $\pm 7\%$
(3) $\pm 4\%$ (4) $\pm 1\%$
- 33.** The resultant of $\vec{A} + \vec{B}$ is $\vec{R}_1$. On reversing the vector $\vec{B}$, the resultant $\vec{R}_2$. What is the value of $R_1^2 + R_2^2$?
- (1) $A^2 + B^2$ (2) $A^2 - B^2$
(3) $2(A^2 + B^2)$ (4) $2(A^2 - B^2)$
- 34.** A particle moves through angular displacement θ on a circular path of radius r . The linear displacement will be :
- (1) $2r \sin (\theta/2)$
(2) $2r \cos (\theta/2)$
(3) $2r \tan (\theta/2)$
(4) $2r \cot (\theta/2)$
- 35.** A vector of length l is turned through the angle θ about its tail. What is the change in the position vector of its head ?
- (1) $l \cos (\theta/2)$
(2) $2l \sin (\theta/2)$
(3) $2l \cos (\theta/2)$
(4) $l \sin (\theta/2)$
- 36.** Given that $A = B$. What is the angle between $\vec{A} + \vec{B}$ and $\vec{A} - \vec{B}$?
- (1) 30° (2) 60°
(3) 90° (4) 180°
- 37.** Given that $\vec{A} + \vec{B} = \vec{R}$ and $\vec{A} + 2\vec{B}$ is perpendicular to $\vec{A}$. Then :
- (1) $2B = R$
(2) $B = 2R$
(3) $B = R$
(4) $B^2 = 2R^2$
- 38.** Error in measurement of radius of a sphere is 1%. Then error in measurement of area is-
- (1) 2% (2) 3%
(3) 4% (4) 5%

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39. If $x = ab$, the maximum percentage error in the measurement of x will be-

(1) $\left(\frac{\Delta a}{a} \times 100\%\right) \times \left(\frac{\Delta b}{b} \times 100\%\right)$

(2) $\left(\frac{\Delta a}{a} \times 100\%\right) \div \left(\frac{\Delta b}{b} \times 100\%\right)$

(3) $\left(\frac{\Delta a}{a} - \frac{\Delta b}{b}\right) \times 100\%$

(4) $\left(\frac{\Delta a}{a} + \frac{\Delta b}{b}\right) \times 100\%$

40. The percentage errors in measurement of mass and speed are 4% and 3% respectively. The error in kinetic energy

will be $[kE = \frac{1}{2}mv^2]$ -

(1) 6%

(2) 7 %

(3) 10%

(4) 12%

41. The number of significant figures in 0.900 is -

(1) 1

(2) 2

(3) 3

(4) 4

42. The velocity of body moving in one-dimension is given by $v = 2s + 3$, where s is displacement. Its acceleration is

(1) $2v$

(2) $4v$

(3) v

(4) $2(2s+3)$

43. The number of significant figures in

(i) 0.03800 and (ii) 90.00 is -

(1) (i) 4 (ii) 4

(2) (i) 2 (ii) 1

(3) (i) 3 (ii) 3

(4) (i) 2 (ii) 4

44. The number of significant figures in 3.04×10^{23} is -

(1) 2

(2) 3

(3) 23

(4) 25

45. The number of significant figures in

0.0060×10^{-18} m is -

(1) 1

(2) 2

(3) 4

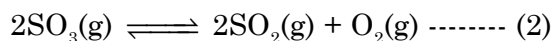
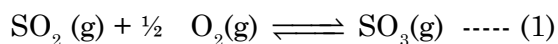
(4) 5

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46. The correct name for the complex $K_3[Fe(C_2O_4)_3]$ is
 (1) Potassium ferric oxalate
 (2) Potassium iron (III) trioxalate
 (3) Potassium trioxalato iron (III)
 (4) Potassium trioxalatoferrate (III).
47. The hybridization involved in $[Ni(CO)_4]$ is
 (1) sp^3 (2) dsp^2
 (3) sp^3d (4) d^2sp^3
48. The K_{sp} for HgS , Ag_2S and PbS are 10^{-31} , 10^{-45} and 10^{-50} respectively. The solubilities are in the order
 (1) $HgS > Ag_2S > PbS$
 (2) $HgS < PbS < Ag_2S$
 (3) $PbS > Ag_2S > HgS$
 (4) $Ag_2S > HgS > PbS$.
49. In a reaction $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ degree of dissociation is 30%. If initial moles of PCl_5 is one then total moles at equilibrium is
 (1) 1.3 (2) 0.7 (3) 1.6 (4) 1.0
50. For reaction $HI \rightleftharpoons \frac{1}{2} H_2 + \frac{1}{2} I_2$ value of K_c is $1/8$ then value of K_c for $H_2 + I_2 \rightleftharpoons 2HI$.
 (1) $\frac{1}{64}$ (2) 64 (3) $\frac{1}{8}$ (4) 8
51. In a equilibrium reaction
 $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
 $\Delta H = -3000$ calories,
 which factor favours dissociation of HI :-
 (1) Low temp. (2) High Pressure
 (3) High temp. (4) Low pressure.
52. $N_2 + 3H_2 \rightleftharpoons 2NH_3$ ($\Delta H = -ve$)
 If temp. of following equilibrium reaction increase then equilibrium state -
 (1) Shift Right side
 (2) Shift left side
 (3) Unchanged
 (4) Nothing say.
53. $C(s) + H_2O(g) \rightleftharpoons H_2(g) + CO(g)$
 by increasing pressure following equilibrium
 (1) Unaffected
 (2) Proceed in backward direction
 (3) Proceed in forward direction
 (4) Unfixed
54. Unit of equilibrium constant K_c for following homogenous reaction :-
 $4NH_3 + 5O_2 \rightleftharpoons 4NO + 6H_2O$ is
 (1) $(Conc)^{-1}$
 (2) $(Conc)^{+1}$
 (3) $(Conc)^{+10}$
 (4) Have no unit
55. Which of the following factor shifted the reaction $PCl_3 + Cl_2 \rightleftharpoons PCl_5$ at left side.
 (1) Adding PCl_5
 (2) Increase pressure
 (3) Constant temp.
 (4) Catalyst.
56. In which of the following process reaction is fastest complete.
 (1) $K = 10$ (2) $K = 1$
 (3) $K = 10^3$ (4) $K = 10^{-2}$

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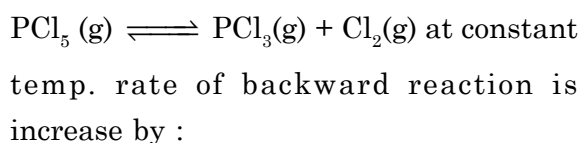
57. At 298 K equilibrium constant K_1 and K_2 of following reaction



The relation between K_1 and K_2 is

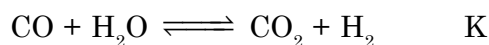
- (1) $K_1 = K_2$ (2) $K_2 = K_1^2$
(3) $K_2 = 1/K_1^2$ (4) $K_2 = 1/K_1$

58. In the following reaction



- (1) Inert gas mixed at constant volume
(2) Cl_2 gas mixed at constant volume
(3) Inert gas mixed at constant pressure
(4) PCl_5 mixed in constant volume.

59. Some gaseous equilibrium are following :



then find out the relation between equilibrium constants :-

- (1) $K = K_1 K_2$
(2) $K = (K_1 K_2)^2$
(3) $K = (K_1 K_2)^{-1/2}$
(4) $K = (K_1/K_2)^{1/2}$

60. For the equilibrium process $x+y \rightleftharpoons xy$. If the concⁿ. of x and y is doubled, then equilibrium constant.

- (1) Become twice
(2) Become half
(3) Unchanged
(4) Become thrice

61. Two moles of PCl_5 were heated in closed vessel of 2 litre capacity, at equilibrium 40% of PCl_5 was dissociated into PCl_3 and Cl_2 . The value of equilibrium constant is :

- (1) 0.267 (2) 0.53
(3) 2.67 (4) 5.3

62. The reaction $\text{A}(\text{g}) + \text{B}(\text{g}) \rightleftharpoons 2\text{C}(\text{g})$ is occur by mixing of 3 moles of A and 1 mole

of B in one litre Container. if α of B is $\frac{1}{3}$,

then K_c for this reaction is :-

- (1) 0.12 (2) 0.25
(3) 0.50 (4) 0.75

63. The reaction $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$ is studied in a one litre vessel at 250°C . The initial concentration of A was $3n$ and that of B was n . When equilibrium was attained, equilibrium concentration of C was found to be equal to the equilibrium concentration of B. What is the concentration of D at equilibrium :

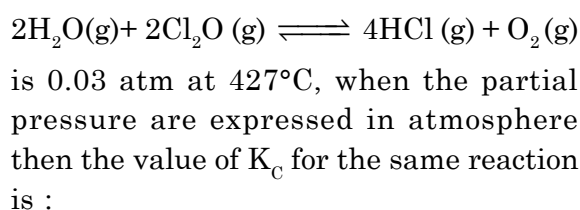
- (1) $n/2$
(2) $(3n - 1/2)$
(3) $(n - n/3)$
(4) n

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64. In a reversible reaction $A \xrightleftharpoons[k_2]{k_1} B$, the initial concentration of A and B are a and b in moles per litre and the equilibrium concentration are $(a - x)$ and $(b + x)$ respectively ; express x in terms of k_1 , k_2 , a and b :

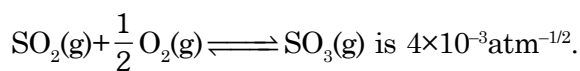
(1) $\frac{k_1 a - k_2 b}{k_1 + k_2}$ (2) $\frac{k_1 a - k_2 b}{k_1 - k_2}$
(3) $\frac{k_1 a - k_2 b}{k_1 k_2}$ (4) $\frac{k_1 a + k_2 b}{k_1 + k_2}$

65. The value of K_p for the reaction



(1) 5.23×10^{-4}
(2) 7.34×10^{-4}
(3) 3.2×10^{-3}
(4) 5.43×10^{-5}

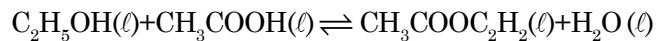
66. The equilibrium constant of the reaction



The equilibrium constant of the reaction $2 SO_3(g) \rightleftharpoons 2 SO_2(g) + O_2(g)$ would be :

(1) 250 atm
(2) 4×10^3 atm
(3) 0.25×10^4 atm
(4) 6.25×10^4 atm

67. When alcohol (C_2H_5OH) and acetic acid are mixed together in equimolar ratio at 27°C, 33% is converted into ester. Then the K_C for the equilibrium :



(1) 4 (2) 1/4
(3) 9 (4) 1/9

68. 2 moles each of SO_3 , CO, SO_2 and CO_2 is taken in a one lit. vessel.

If K_C for $SO_3 + CO \rightleftharpoons SO_2 + CO_2$ is 1/9 then :

(1) total no. of moles at equilibrium are less than 8
(2) $n(SO_3) + n(CO_2) = 4$
(3) $[n(SO_2)/n(CO)] < 1$
(4) both (2) and (3)

69. An equilibrium mixture in a vessel of capacity 100 litre contain 1 mol N_2 , 2 mol O_2 and 3 mol NO. No. of moles of O_2 to be added so that at new equilibrium the conc. of NO is found to be 0.04 mol/lit :

(1) (101/18) (2) (101/9)
(3) (202/9) (4) None of these

70. Which of the following expressions is/are not true ?

(1) $[H^+] = [OH^-] = \sqrt{K_w}$ for a neutral solution at all temperatures.
(2) $[H^+] > \sqrt{K_w}$ & $[OH^-] < \sqrt{K_w}$ for an acidic solution
(3) $[H^+] < \sqrt{K_w}$ & $[OH^-] > \sqrt{K_w}$ for an alkaline solution
(4) $[H^+] = [OH^-] = 10^{-7} \text{ M}$ for a neutral solution at all temperatures

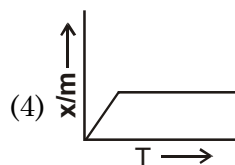
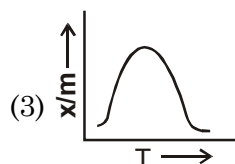
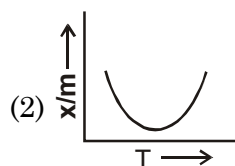
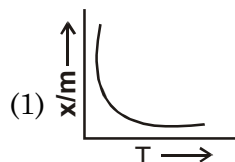
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71. An acid solution of pH 6 is diluted thousand times. The pH of solution becomes approx-
(1) 6.96 (2) 6
(3) 4 (4) 9
72. Which of the following solution will have a pH exactly equal to 8 ?
(1) 10^{-8} M HCl solution at 25°C
(2) 10^{-8} M H^{+} solution at 25°
(3) 2×10^{-6} M $\text{Ba}(\text{OH})_2$ solution at 25°C
(4) 10^{-6} M NaOH solution at 50°C
73. The number of hydrogen ions in 10 mL of a solution with pH = 13 is-
(1) 10^{13}
(2) 6.023×10^8
(3) 6.023×10^{13}
(4) 6.023×10^{10}
74. A 50 mL solution of pH = 1 is mixed with a 50 mL solution of pH = 2. then pH of the mixture will be nearly-
(1) 0.76 (2) 1.26
(3) 1.76 (4) 2.26
75. The pH of a solution obtained by mixing 50 mL of 0.4 N HCl and 50 mL of 0.2 N NaOH is-
(1) $-\log 2$ (2) $-\log 0.2$
(3) 1.0 (4) 2.0
76. The pH of a solution is 7.00. To this solution sufficient base is added to increase the pH to 12.0. The increase OH^{-} ion concentration is-
(1) 5 times
(2) 1000 times
(3) 10^5 times
(4) 4 times
77. Which of the following solution will have pH close to 1.0 ?
(1) 100 mL of M/10 HCl + 100 mL of M/10 NaOH
(2) 55 mL of M/10 HCl + 45 mL of M/10 NaOH
(3) 10 mL of M/10 HCl + 90 mL of M/10 NaOH
(4) 75 mL of M/5 HCl + 25 mL of M/5 NaOH.
78. 50% neutralization of a solution of formic acid ($K_a = 2 \times 10^{-4}$) with NaOH would result in a solution having a hydrogen ion concentration of-
(1) 2×10^{-4} (2) 3.7
(3) 2.7 (4) 1.85
79. The pH of a solution obtained by mixing 100 mL of 0.2 M CH_3COOH with 100 mL of 0.2 M NaOH would be ($\text{p}K_a$ for $\text{CH}_3\text{COOH} = 4.74$ and $\log 2 = 0.301$)
(1) 4.74 (2) 8.87
(3) 9.10 (4) 8.57
80. The solubility of $\text{Fe}(\text{OH})_3$ would be maximum in-
(1) 0.1 M NaOH
(2) 0.1 M HCl
(3) 0.1 M KOH
(4) 0.1 M H_2SO_4
81. Arrange in increasing order of solubility of AgBr in solutions given :
(i) 0.1 M NH_3
(ii) 0.1 M AgNO_3
(iii) 0.2 M NaBr
(iv) pure water
(1) (iii) < (ii) < (iv) < (i)
(2) (iii) < (ii) < (i) < (iv)
(3) (iii) < (ii) = (i) < (iv)
(4) (ii) < (iii) < (iv) < (i)

SPACE FOR ROUGH WORK

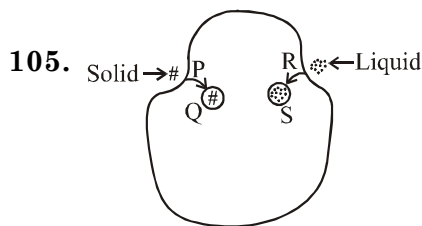
82. The dispersion medium in Gel is -
 (1) Solid
 (2) Liquid
 (3) Water
 (4) Gas
83. Liquid-Liquid sols. are known as -
 (1) Aerosols
 (2) Emulsions
 (3) Foam
 (4) Gels
84. Which of the following metal sols. cannot be prepared by Bredig's arc method -
 (1) Copper
 (2) Potassium
 (3) Gold
 (4) Platinum
85. The size of colloidal particles is in between-
 (1) $10^{-7} - 10^{-9}$ cm
 (2) $10^{-9} - 10^{-11}$ cm
 (3) $10^{-5} - 10^{-7}$ cm
 (4) $10^{-2} - 10^{-3}$ cm
86. As_2S_3 sol. is -
 (1) Positive colloid
 (2) Negative colloid
 (3) Neutral colloid
 (4) None of the above
87. Point out the false statement -
 (1) Colloidal sols are homogenous
 (2) Colloids carry +ve or -ve charge
 (3) Colloids show tyndall effect
 (4) The size range of colloidal particles is $10 - 1000 \text{ \AA}$

88. A metal sulphide sol. can be flocculated, by one of the following -
 (1) $[Fe(CN)_6]^{4-}$
 (2) SO_4^{-2}
 (3) $BeCl_2$
 (4) Al^{3+}
89. Which of the following has largest protecting power -
 (1) Gelatin (Gol no. = 0.01)
 (2) Dextrin (Gol no. = 15)
 (3) Potato starch (Gol no. = 25)
 (4) Albumin (Gol no = 0.25)
90. Which plot is the adsorption isobar for chemisorption where x is the amount of gas adsorbed on mass m (at constant pressure) at temperature T -



SPACE FOR ROUGH WORK

- 91.** Which of the following plastid store protein?
(1) Aleuroplast (2) Amyloplast
(3) Chromoplast (4) Elaioplast
- 92.** Mitochondria and chloroplasts are :-
(1) Considered as part of the endomembrane system
(2) Single membrane bound cell organelles
(3) Divide by binary fission
(4) Having 80 S ribosomes
- 93.** Ribosomes of mitochondria, chloroplast and R.E.R are respectively :-
(1) 80S, 70S, 70S
(2) 80S, 80S, 80S
(3) 70S, 70S, 80S
(4) 70S, 80S, 70S
- 94.** Which of the following has direct connections to outer membrane of nuclear membrane?
(1) RER (2) Golgi bodies
(3) Lysosomes (4) Mitochondria
- 95.** Plasma membrane is regarded as asymmetric because :-
(1) Carbohydrate in outer and spectrin protein inner sides are present
(2) It contains carbohydrate
(3) Protein icebergs in sea of lipid
(4) Made up of glycoprotein and lipid
- 96.** Choose the incorrect match
(1) Cristae – Infolding of inner membrane of mitochondria
(2) Granna – Stacks of thylakoids
(3) Ribosome – Protein synthesis
(4) Elaioplasts – Store protein granules
- 97.** In human being, the RBC membrane has approximately :-
(1) 52% protein, 40% lipid
(2) 40% protein, 52% lipid
(3) 42% protein, 50% lipid
(4) 30% protein, 50% lipid
- 98.** Choose correct one w.r.t. composition of algae cell wall :-
(1) Cellulose + Galactans + Mannans + Minerals like CaCO_3
(2) Hemicellulose and pectin
(3) Cellulose + Hemicellulose + Pectin + Minerals like CaCO_2
(4) Suberin + Cellulose + Galactans + Silica
- 99.** Cathepsin enzyme is helpful in digestion of tail of tadpole larva, it is secreted by :-
(1) Sphaerosome
(2) Peroxysome
(3) Lysosome
(4) Glyoxysome
- 100.** The principle protein of Cilia and Flagella is :-
(1) Tubulin (2) Albumin
(3) Globulin (4) Fibrinogen
- 101.** Function of centriole is :-
(1) Formation of spindle fibre
(2) Formation of Nucleolus
(3) Initiation of cell membrane
(4) Formation of cell plate
- 102.** The site of synthesis and degradation of H_2O_2 in a plant cell is :-
(1) Lysosome (2) Sphaerosome
(3) Peroxisome (4) Microsome
- 103.** Microtubules are present in all, except :-
(1) Flagella
(2) Spindle fibres
(3) Centrioles
(4) Microbodies
- 104.** An elaborate network of filamentous and proteinaceous structures present in the cytoplasm of an eukaryotic cell is called :-
(1) Cytoskeleton
(2) Chromatin fibres
(3) Polysome
(4) Microsome

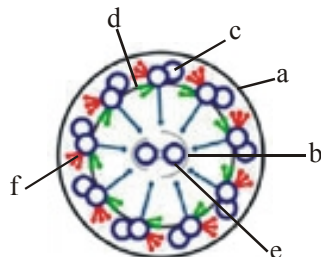


Identify P, Q, R and S in above given diagram. Which is showing ingestion of food material?

Option :-

- (1) P → Pinocytosis, Q → Phagosome, R → Phagocytosis, S → Pinosome
- (2) P → Phagocytosis, Q → Phagosome, R → Pinocytosis, S → Pinosome
- (3) P → Pinocytosis, Q → Pinosome, R → Phagocytosis, S → Phagosome
- (4) P → Pinosome, Q → Pinocytosis, R → Phagosome, S → Phagocytosis

106. Identify the figure and find the correct option

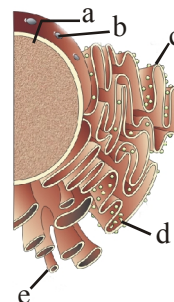


- (1) a-cell membrane, e- interdoubt bridge, c- peripheral doublet, f-dynein arms
- (2) c-peripheral doublet e-linker, d-nexin, a-cell membrane
- (3) a-cell membrane, c-linker, d-dynein arms, f-radial spokes
- (4) b-central sheath, c- peripheral doublet, e-central singlet, f-dynein arm

107. _____ is present in germinating seeds and concerned with oxidation of fats

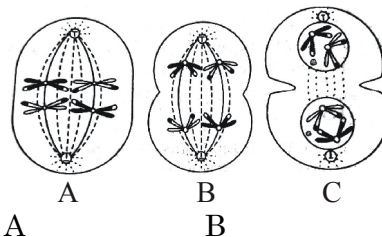
- (1) Peroxisomes
- (2) Glyoxysomes
- (3) Sphaerosomes
- (4) Lomasomes

108. Identify the figure and find the correct match



- (1) a -nucleus, b -nucleopore, c -RER, d -ribosomes, e-SER
- (2) a -nucleus, b - nucleopore, c - SER, d - ribosomes, e-RER
- (3) a-nucleolus, b -nucleopore, c -RER, d -ribosomes, e-SER
- (4) a -nucleolus, b -nucleopore, c -SER, d -ribosomes, e-RER

109. Identify the given figures showing meiotic phases and select the correct option.



- (1) Metaphase Anaphase Telophase
- (2) Metaphase-I Anaphase-I Telophase-I
- (3) Metaphase-II Anaphase-II Telophase-II
- (4) Anaphase-I Metaphase-I Telophase-I

110. Match the following columns.

Column I

Column II

- | | |
|----------------------------|----------------|
| A. Terminalisation | 1. Metaphase-I |
| B. Histone synthesis | 2. Anaphase-I |
| C. Disjunction | 3. Diakinesis |
| D. Interkinesis | 4. Meiosis-II |
| E. Double equatorial plate | 5. S-phase |

Codes

- | | A | B | C | D | E |
|-----|---|---|---|---|---|
| (1) | 5 | 2 | 1 | 3 | 4 |
| (2) | 3 | 5 | 4 | 2 | 1 |
| (3) | 2 | 5 | 4 | 1 | 3 |
| (4) | 3 | 5 | 2 | 4 | 1 |

111. Crossing over occurs during :-

- (1) Diplotene (2) Pachytene
(3) Zygotene (4) Leptotene

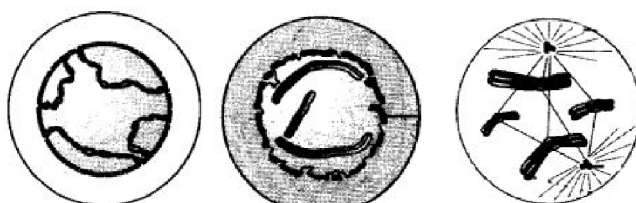
112. In Meiosis the homologous chromosomes are separated during :-

- (1) Metaphase I (2) Prophase II
(3) Anaphase I (4) Anaphase II

113. Chromosome number is reduced to half in :-

- (1) Equational division
(2) Meiosis II
(3) Homotypic division
(4) Meiosis I

114. Recognise the figure and find out the correct matching :-

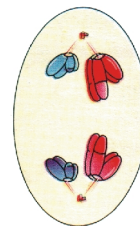


- (a) (b) (c)
- (1) a-Early prophase, b-late prophase, c-transition to metaphase
(2) b-Early prophase, c-late prophase, a-transition to metaphase
(3) c-Early prophase, a-late prophase, b-transition to metaphase
(4) b-Early prophase, a-late prophase, c-transition to metaphase

115. At which stage of meiosis, do these events occurs ?

	Bivalents more clearly visible	Formation of synaptonemal complex	Terminalisation of chiasmata
(1)	Leptotene	Zygotene	Pachytene
(2)	Zygotene	Pachytene	Diakinesis
(3)	Pachytene	Zygotene	Diakinesis
(4)	Zygotene	Diplotene	Diakinesis

116. Identify the below stage with their respective event :-



- (1) Anaphase-II Splitting of the centromere
(2) Anaphase-I Homologous chromosome separate
(3) Anaphase-I Splitting of the centromere
(4) Metaphase-I Bivalent chromosomes align on the equatorial plate.

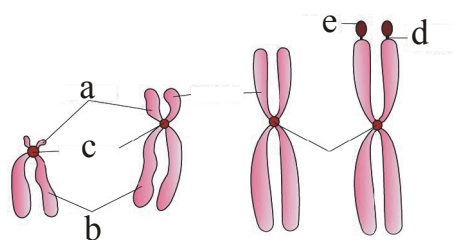
117. Match column-I with column-II and select the **correct** option from the codes given below.

Column A		Column B	
A.	NOR	I	Secondary constriction
B.	Nuclear lamina	II	MTOC
C.	Histone	III	Intermediate filament
D.	Centrosome	IV	Nucleosome

Options :-

- (1) A-I, B-III, C-IV, D-II
(2) A-I, B-III, C-II, D-IV
(3) A-III, B-I, C-IV, D-II
(4) A-I, B-IV, C-II, D-III

118. Recognize the figure and find the correct match



- (1) a-p arm, b-long arm, c-primary constriction, d-SAT
(2) a-short arm, b-q arm, c-centromere, d-secondary constriction
(3) a-q arm, b-p arm, c-centromere, d-satellite
(4) a-p arm, b-q arm, c-primary constriction, e-secondary constriction

- 119.** A structural and functional unit of nature where living organisms interact among themselves and also with the surrounding physical environment is ecosystem. Which of the following is not correct for ecosystem?
- (1) Term ecosystem was given by A.G. Tansley
 - (2) It is self regulatory and self sustaining unit
 - (3) It cannot maintain resilience
 - (4) Regular input of energy is essential to maintain homeostasis
- 120.** The structural component of ecosystem includes :-
- (1) Species composition
 - (2) Stratification
 - (3) Decomposition
 - (4) Both (1) and (2)
- 121.** Stratification means vertical distribution of :-
- (1) Individuals of different species in temperate needle like forest
 - (2) Individual of same species
 - (3) Trees, shrubs, and herbs
 - (4) Bacteria, cyanobacteria and fungi
- 122.** Species having fewer number of individuals but determine the ability of other species to exist and removal of which affect large number of species in an ecosystem is :-
- (1) Keystone species
 - (2) Critical link species
 - (3) Edge species
 - (4) Ecotone
- 123.** The annual net primary productivity of the whole biosphere approximately is :-
- (1) 170 billion tons
 - (2) 80 billion tons
 - (3) 55 billion tons
 - (4) 55 million tons
- 124.** The rate of formation of new organic matter by consumers is known as :-
- (1) GPP
 - (2) NPP
 - (3) Secondary Productivity
 - (4) Net production efficiency
- 125.** Nutrient immobilisation is :-
- (1) Incorporation of nutrients in living micro organisms
 - (2) Covalent linking of nutrients with one another
 - (3) Loss of nutrients during fragmentation
 - (4) Loss of nutrients during soil erosion
- 126.** Based on the source of their nutrition of food organism occupy a specific position in the food chain, that is known as their :-
- (1) Trophic level
 - (2) Niche
 - (3) Habitat
 - (4) Micro climate
- 127.** Plant capture only% of the PAR.
- (1) 1-5
 - (2) 25
 - (3) 2-10
 - (4) 1.6-5
- 128.** Which one of the following pyramid would be inverted?
- (1) Pyramid of number in a grassland ecosystem
 - (2) Pyramid of number in a tree ecosystem
 - (3) Pyramid of Biomass in sea
 - (4) More than one option is correct

129. The first biotic community which develops in a bare area is called pioneer community which is characterised by :-

- (i) High productivity
- (ii) Low productivity
- (iii) High multiplication
- (iv) Low multiplication

- (1) (i), (iii)
- (2) (i), (iv)
- (3) (ii), (iii)
- (4) (ii), (iv)

130. Primary succession takes comparatively longer time to achieve climax because :-

- (1) Soil is not formed in the beginning
- (2) Soil has only some propagules
- (3) Soil has excess of minerals
- (4) Soil has low humus content

131. Climax community differs from pioneer community in

- (1) Increase in structural complexity
- (2) Increase in species diversity
- (3) Niche specialisation
- (4) All are correct

132. What is pioneer community for hydrarch and xerarch succession respectively?

- (1) Zooplanktons and mosses
- (2) Lichens and phytoplanktons
- (3) Submerged plants and mosses
- (4) Phytoplanktons and lichens

133. How many of given ecosystem are natural and man-made ecosystem respectively?

Forest, Grassland, Crop field, River, Estuary, Fishery tank, Aquaria, Lakes

- (1) 5, 3
- (2) 3, 5
- (3) 4, 4
- (4) 6, 2

134. Select the correct sequences of stages in xerarch succession.

- (1) Lichens → Mosses → Perennial grasses → Annual grasses → Shrubs → Forests
- (2) Lichens → Mosses → Annual grasses → Perennial grasses → Shrubs → Forests
- (3) Mosses → Lichens → Perennial grasses → Annual grasses → Shrubs → Forests
- (4) Mosses → Lichens → Shrubs → Annual Grasses → Perennial grasses → Shrubs → Forests

135. Arrange various stages of primary succession in a newly formed fresh water pond are given :-

- (A) Scrub stage
- (B) Sub merged free floating plant stage
- (C) Marsh meadow stage
- (D) Reed swamp stage
- (E) Phytoplanktons
- (F) Submerged plant stage
- (G) Forest

- (1) E, F, B, C, D, A, G
- (2) E, B, F, D, C, A, G
- (3) E, F, B, D, C, A, G
- (4) E, B, C, D, B, A, G

136. Find out the correct match from the following table :-

	Column-I	Column-II	Column-III
A	Goblet cells	Mucus	Prevent mucosa layer from damage of HCl
B	Lysozyme	Saliva	Antibacterial agent
C	Saliva	Subparotid gland	Secrete salivary amylase
D	Oxyntic cells	HCl	Stimulate gastric lipase

- (1) Only 'A' (2) A and B
(3) Only C (4) C and D

137. Consider the following sequences ('a' to 'd') of histological layers of alimentary canal and select the answer from the options given below. :-

- (a) serosa, muscularis, submucosa, mucosa
(b) serosa, submucosa, muscularis, mucosa
(c) mucosa, submucosa, muscularis, serosa
(d) mucosa, muscularis, submucosa, serosa
(1) sequence a and d (2) sequence b and c
(3) sequence a and c (4) sequence b and d

138. Consider the following four statements (a-d) and select the option which includes all the correct ones only :-

- (a) the stomach stores food for 30 minutes only
(b) Intestinal juice is also called succus entericus
(c) Mixture of food with saliva is called bolus
(d) Pancreatic juice contain digestive enzymes for digestion of proteins only.
(1) Statements a,c,d
(2) Statements b,c
(3) Statements a,b,d
(4) Statement d only

139. Brunner's glands are characterstics of :-

- (1) Submucosa of duodenum
(2) Mucosa of jejunum
(3) Submucosa of stomach
(4) Mucosa of ileum

140. In pernicious anaemia, failure of erythrocytes to mature is mainly due to lack of :-

- (1) Vitamin-K
(2) Thiamine
(3) Pyridoxine
(4) Cyanocobalamine

141. Milk protein casein $\xrightarrow{(y)}$ paracasein
 $\xrightarrow{(M)}$ calcium paracaseinate (curding of milk) :-

In the above reaction, latter (y) and (M) denote:-

- (1) Rennin and Ca^{+2} respectively
(2) Ca^{+2} and rennin respectively
(3) Renin, HCl and Ca^{+2} respectively
(4) Amylase and Ca^{+2} respectively

142. If for some reason the parietal cells of the stomach epithelium become partially non-functional what is likely to happen ?

- (1) The pancreatic enzymes and specially the trypsin and lipase will not work efficiently
(2) The pH of stomach will fall abruptly
(3) Steapsins will be more effective
(4) Proteins will not be adequately hydrolysed by pepsin into proteoses and peptones

143. Consider the following four statements (a-d) and select the option which includes all the correct ones only :-

- (a) Serosa is outermost layer and is made up by mesothelium and connective tissue
- (b) Submucosa layer is made up of soft and areolar connective tissue, blood vessels, lymph vessels and nerve fibres.
- (c) Inner most layer of alimentary canal is made up by mucosa which forms the folds and ridges
- (d) Muscularis layer is made up by outer circular muscles and inner longitudinal muscles.

Options :-

- (1) Statements (b), (c) & (d)
- (2) Statements (a) and (d)
- (3) Statements (a), (b) & (c)
- (4) Statements (d) only

144. Consider the following four statements (a-d) and select the option which includes all the correct ones only.

- (a) On the dorsal surface of tongue, taste papillae are found and all papillae contain taste buds.
- (b) Opening of stomach into duodenum is guarded by pyloric sphincter.
- (c) Opening of oesophagus into stomach is guarded by cardiac sphincter.
- (d) Dorsal surface of tongue is connected with base of buccal cavity through frenulum linguae.

Options :-

- (1) Statements (a) and (b)
- (2) Statements (a), (b) & (c)
- (3) Statements (b) & (c)
- (4) Statements (b), (c) & (d)

145. Which one is correct dental formula of adult human ?

- (1) $I \frac{1}{1} C \frac{2}{2} Pm \frac{2}{2} M \frac{3}{3} \times 2$
- (2) $I \frac{2}{2} C \frac{1}{1} Pm \frac{2}{2} M \frac{3}{3} \times 2$
- (3) $I \frac{2}{2} C \frac{0}{0} Pm \frac{3}{3} M \frac{2}{2} \times 2$
- (4) $I \frac{2}{2} C \frac{1}{1} Pm \frac{3}{3} M \frac{2}{2} \times 2$

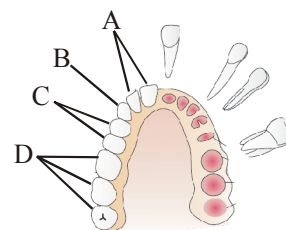
146. Find out the correct match from the following table :-

	Column-I	Column-II	Column-III
(i)	Ptyalin enzyme	Salivary gland	Protein digesting enzyme
(ii)	Pepsin	Gastric gland	Protein digesting enzyme
(iii)	Amylase	Pancreas	Fat digesting enzyme

Options :-

- (1) Only (i)
- (2) (i) & (ii) both
- (3) Only (ii)
- (4) (ii) & (iii) both

147. An arrangement of teeth on jaw bone in human beings are shown in given below diagram and different types of four teeth are labelled by alphabets A, B, C and D. Which one of these is not correctly identified in the option, given along with its structure and function ?



	Teeth	Structure	Function
(1) B	Canines	Long & pinpointed	Tearing & shearing
(2) C	Premolars	Rectangular	Gnawing
(3) D	Molars	Rectangular	Chewing
(4) A	Incisors	Chisel shaped	Cutting / gnawing

148. Find out the correct match from the following table :-

	<i>Column-I Name of salivary gland</i>	<i>Column-II Their duct</i>	<i>Column-III Their property</i>
(i)	<i>Parotid gland</i>	<i>Stenson's duct</i>	<i>Largest salivary gland</i>
(ii)	<i>Sublingual gland</i>	<i>Wirsung duct</i>	<i>Smallest salivary gland</i>
(iii)	<i>Submandibular gland</i>	<i>Wharton's duct</i>	<i>Maximum salivation</i>

Options :-

- (1) Only (ii)
- (2) (i) and (iii)
- (3) Only (i)
- (4) (i), (ii) and (iii) all

149. Read the following statements :-

- (i) Three muscular cords are present in wall of colon are known as Teniae coli
- (ii) Haustra are the autopouching and helped in increasement of absorptive surface area in ileum
- (iii) "Folds of kerkring" are present in small intestine and helped in increasing of absorptive surface area
- (iv) Ileo-caecal valve is present between colon and rectum and it regulates entry of undigested food materials into rectum.

Out of these which statements are incorrect from followings ?

- (1) (i) & (ii) (2) (ii) & (iii)
- (3) (i) & (iv) (4) (ii) & (iv)

150. "Enamel" is originated by (A) and secreted by (B) cells. Select the option in which (A) and (B) are correctly identified ?

- (1) A = Mesoderm; B = Odontoblast
- (2) A = Ectoderm ; B = Ameloblast
- (3) A = Mesoderm; B = Osteoclast
- (4) A = Ectoderm; B = Osteoblast

151. The utilisation of nutrients by the tissues for their activities is called :-

- (1) Digestion
- (2) Absorption
- (3) Assimilation
- (4) Egestion

152. Which of the following layer of alimentary canal is highly vascular in nature :

- (1) Serosa
- (2) Muscularis
- (3) Submucosa
- (4) Mucosa

153. The stomach stores the food for :

- (1) 1 - 2 hrs. (2) 2 - 3 hrs.
- (3) 3 - 4 hrs. (4) 4 - 5 hrs.

154. Which one of the following is not a cell of gastric glands :

- (1) Mucus neck cells
- (2) Paneth cells
- (3) Peptic cells
- (4) Parietal cells

155. Which component of food is hydrolysed in oral cavity by the action of salivary amylase ?

- (1) fat
- (2) carbohydrate
- (3) protein
- (4) all of these

156. Match the Column I with Column II and choose the correct option :

Column - I	Column - II
(a) Bilirubin and Biliverdin	(i) Parotid
(b) Hydrolysis of starch	(ii) Bile
(c) Digestion of fat	(iii) Lipases
(d) Salivary gland	(iv) Amylases
(1) a = (ii), b = (iv), c = (iii), d = (i)	
(2) a = (i), b = (ii), c = (iii), d = (iv)	
(3) a = (iv), b = (iii), c = (ii), d = (i)	
(4) a = (i), b = (iv), c = (iii), d = (ii)	

157. How many human teeth appear twice during the life span of an individual :

- (1) 12 (2) 20
- (3) 28 (4) 32

158. Jaundice may be caused by retarded function of :

- (1) Stomach (2) Liver
(3) Intestines (4) Pancreas

159. Match the Column I with Column II and choose the correct option :

Column I	Column II
A. Goblet cells	1. Antibacterial agent
B. Lysozymes	2. Mucus
C. Saliva	3. HCl
D. Oxyntic cells	4. Sublingual gland

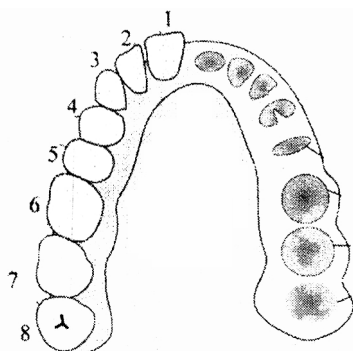
(1) A = 3, B = 1, C = 4, D = 2

(2) A = 1, B = 3, C = 4, D = 2

(3) A = 2, B = 3, C = 1, D = 4

(4) A = 2, B = 1, C = 4, D = 3

160. The diagram of human jaw is given showing the teeth. The names of teeth numbered 3 and 4 are :-



- (1) Incisor and Canine
(2) Canine and Premolar
(3) Premolar and molar
(4) Incisor and Premolar

161. Which of the following is not a disorder of digestive system :

- (1) Jaundice (2) Vomiting
(3) Angina pectoris (4) Constipation

162. Bile is formed in :

- (1) Gall bladder (2) Liver
(3) Spleen (4) Blood

163. Proteolytic enzymes do not corrode lining of alimentary canal as :

- (1) They are secreted in inactive form
(2) Lining layer of alimentary canal does not contain protein
(3) The enzymes are not capable of digesting fat
(4) None of these

164. Where do certain symbiotic microorganism normally occurs in human body ?

- (1) Vermiform appendix and rectum
(2) Duodenum
(3) Caecum
(4) Oral lining and tongue surface

165. The distal part of the stomach that opens into duodenum is called :

- (1) Fundus (2) Pylorus
(3) Omentum (4) Cardiac

166. Which of the following is the wrong match in the given table?

Options :

		Parotid gland	Submaxillary gland	Sublingual gland
1	Location	Below ear	Jaw angle	Below tongue
2	Number	One pair	One pair	One pair
3	Duct	Stenson duct	Duct of Rivinus	Wharton duct
4	Secretion	Saliva + enzyme	Saliva	Saliva

167. Which part of alimentary canal does not secrete any digestive enzyme ?

- (1) Mouth (2) Stomach
(3) Duodenum (4) Oesophagus

168. The epithelial cells lining the stomach of vertebrates is protected from the damage by HCl because :

- (1) HCl is too dilute
(2) HCl is neutralized by stomach
(3) The epithelial cells are resistant to the action of HCl
(4) The epithelial cells are covered by a mucous secretion.

169. Pulp cavity of teeth is surrounded by :

- (1) Enamel (2) Gums
(3) Bones (4) Odontoblasts

170. Which one of these are most essential for the body growth and formation of new cells ?

- (1) Fats (2) Sugars
(3) Vitamins (4) Proteins

- 171.** Peyer's patches are found in
- (1) Lamina propria of ileum
 - (2) submucosa of duodenum
 - (3) Serosa of jejunum
 - (4) Muscularis of stomach
- 172.** 'Every cell of the body contributes gemmulate to the germ cells and so shares in the transmission of inherited characters'. This theory is known as
- (1) theory of inheritance of acquired characteristics
 - (2) theory of germplasm
 - (3) theory of pangenesis
 - (4) theory of mutations
- 173.** The next stage in the evolution of man after Australopithecus was :
- (1) *Homo habilis*
 - (2) *Homo sapiens*
 - (3) *Homo erectus*
 - (4) *Homo pekinensis*
- 174.** If a particular animal has shelled eggs, hair and teats on the body and has cloaca, it may be a connecting link between
- (1) Reptiles and birds
 - (2) Birds and mammals
 - (3) Reptiles and mammals
 - (4) None of these
- 175.** Presence of coelacanth fish was observed in
- (1) South America
 - (2) South Africa
 - (3) North America
 - (4) North Africa

- 176.** Identify the below diagram.



- (1) Brachiosaurus
 - (2) Stegosaurus
 - (3) Tasmanian wolf
 - (4) Banded antheater
- 177.** Which of the following evidences does not favour the Lamarckian concept ?
- (1) Absence of limbs in snakes
 - (2) Presence of webbed toes in aquatic birds
 - (3) Melanization in peppered moth in industrial area
 - (4) Lack of pigment in cave dwelling animals
- 178.** Most recent man found as fossil was
- (1) Java man
 - (2) Peking man
 - (3) Cromagnon man
 - (4) Hiedelberg man
- 179.** Match the column
- | | | | |
|-----|------------------|-----|---|
| I | Australopithecus | i | First human like |
| II | Homo Habilis | ii | Used hides to protect the body |
| III | Homo Erectus | iii | Arose in Africa |
| IV | Neanderthal man | iv | Probably ate meat |
| V | Homo Sapiens | v | Hunted with stone weapons and at fruits |
- (1) I-v, I-ii, III-iv, IV-i, V-iii
 - (2) I-i, I-ii, III-iv, IV-iii, V-v
 - (3) I-v, I-i, III-iv, IV-ii, V-iii
 - (4) I-v, I-ii, III-ii, IV-i, V-iv
- 180.** Darwin had made his sea voyage through
- (1) H.M.S. Beagle
 - (2) Titanic
 - (3) Carpethiya
 - (4) Rocky Mountain

SPACE FOR ROUGH WORK

Result : NEET (UG) 2019

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AIR

1



NALIN KHANDELWAL
Classroom

✓ **8 in Top 10 All India Ranks**

✓ **61 in Top 100 All India Ranks**

TOTAL STUDENTS QUALIFIED

Classroom
63415

95102*

Distance
31687

AIR 2  BHAVIK BANSAL Distance	AIR 4  SWASTIK BHATIA Distance	AIR 5  ANANT JAIN Classroom	AIR 6  SARTHAK R. BHAT Distance	AIR 8  DHRUV KUSHWAHA Distance	AIR 9  MIHIR RAI Distance	AIR 10  RAGHAV DUBEY Classroom									
AIR-11  HEMANT KHANDELA Distance	AIR-17  RIDHAM AGARWAL Distance	AIR-18  HARSHVI N. J. Classroom	AIR-19  ARUNANGSHU B. Classroom	AIR-20  SATVIK LODHA Classroom	AIR-21  HARSH JAIN Classroom	AIR-22  HARDIK GUPTA Classroom	AIR-24  LALIT AGARWAL Classroom	AIR-26  APOORV RAGHAV Classroom	AIR-27  PARTH GILL Distance	AIR-28  GAUTAM CHELLANI Distance	AIR-30  HARSH AGARWAL Classroom	AIR-37  LOHITASHVA GOYAL Distance	AIR-38  VISHWA RAKESH Distance	AIR-39  ANURAG GAUTAM Distance	AIR-44  M. H. SAYAN SHAH Classroom
AIR-45  SHIKHAR GUPTA Distance	AIR-46  DIPRASON DAS Distance	AIR-47  BAHADUR SINGH Distance	AIR-48  AMAN AGRAWAL Classroom	AIR-49  VARUN CHOPRA Distance	AIR-51  ISHAAN MISHRA Distance	AIR-52  DISHA S. AGRAWAL Distance	AIR-53  SARVAGYA JAIN Distance	AIR-54  HARRY GOYAL Distance	AIR-56  ISHIKA GUPTA Distance	AIR-58  SHREYAS AGRAWAL Classroom	AIR-60  ABHAY MITTAL Classroom	AIR-61  CHIRAG JORVEKAR Distance	AIR-63  VINEET HARSH Distance	AIR-65  VAMSHI K. NEELA Distance	AIR-66  SHREY MUDGAL Classroom
AIR-69  PRIYANSHU AGARWAL Classroom	AIR-70  ISHIKA KAMRA Classroom	AIR-71  STUTI KHANDWALA Classroom	AIR-72  NAVEEN SHARMA Distance	AIR-75  SARTHAK SHARMA Classroom	AIR-76  VANSHIKA ARORA Distance	AIR-79  SAIYAM V. BHATT Classroom	AIR-81  UDITH MOHAN Distance	AIR-84  PRAJNAPAN BASU Distance	AIR-85  HARSH RAMOLIYA Distance	AIR-86  UTKARSH SHARMA Classroom	AIR-87  NISHITA Distance	AIR-89  ARYAN BANSAL Distance	AIR-91  SANGEETA AGRAWAL Classroom	AIR-92  YASHDEEP Classroom	AIR-93  ISHAN GUPTA Classroom
AIR-94  MEGHA GUPTA Classroom	AIR-96  GAUTAM KAUSHAL Classroom	AIR-97  POOJA YADAV Distance											AIR-99  PRAGYA MITRA Distance	AIR-100  VIVEK KUMAR Classroom	



Authenticity of Result :
Power of **ALLEN**



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Congratulations!

*As Per Result Compiled so far